

Translation



The following guide announces China's funding priorities for explainable and generalizable AI basic research in 2025. The Chinese government is funding cutting-edge basic research on various approaches to AI, including generative large models but also brain-inspired and cognitive AI models. The guide also encourages and subsidizes novel applications of AI in physical modeling, biological and health research, and materials science.

Title

Guide to the 2025 Annual Projects for the Major Research Program on Explainable and Generalizable Next-Generation Artificial Intelligence Methods
可解释、可通用的下一代人工智能方法 重大研究计划2025年度项目指南

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National Natural Science Foundation of China (NSFC; 国家自然科学基金委员会). NSFC is an organization under the Ministry of Science and Technology (MOST; 科学技术部; 科技部) that oversees and audits Chinese scientific research funds, principally those that support basic research.

Source

NSFC website, January 24, 2025.

The Chinese source text is available online at:

<https://www.nsf.gov.cn/publish/portal0/tab442/info94367.htm>

An archived version of the Chinese source text is available online at: <https://perma.cc/EN2E-HLP8>

U.S. \$1 ≈ 7.2 Chinese Yuan Renminbi (RMB), as of March 24, 2025.

Translation Date

March 24, 2025

Translator

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Editor

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Guide to the 2025 Annual Projects for the Major Research Program on Explainable and Generalizable Next-Generation Artificial Intelligence Methods

The Major Research Program on Explainable¹ and Generalizable Next-Generation Artificial Intelligence Methods (“the Major Research Program”) is geared towards: The major national strategic requirements of artificial intelligence (AI) development, taking the basic scientific problems of AI as the core; developing a system of new AI methods; and promoting China’s basic AI research and talent cultivation, thereby supporting China's dominant (主导) position in the new round of international competition in

¹ Translator’s note: In the context of AI, the Chinese word 可解释性 can be translated as either “explainable” or “interpretable.” This translation opts for the former.

science and technology (S&T).

I. Scientific Objectives

The Major Research Program is geared towards: Poor robustness, poor explainability, strong data dependence, and other basic scientific problems in AI methods as represented by deep learning; uncovering the basic principles of machine learning; and developing explainable and generalizable next-generation AI methods, so as to promote the innovation and application of AI methods in scientific fields.

II. Core Scientific Problems

The Major Research Program is aimed at the basic scientific problems of explainable and generalizable next-generation AI methods, centered around conducting research on the following three core scientific problems.

(i) Basic principles of deep learning.

Deeply uncover the dependency of deep learning on hyperparameters, understand the working principles behind deep learning, and, for deep learning methods, establish approximation theory, generalization error analysis theory, and the convergence theory of optimization algorithms.

(ii) Explainable and generalizable next-generation AI methods.

By combining rules and learning, establish new AI methods that are high-precision, explainable, and generalizable, and do not rely on large amounts of annotated data. Develop the databases and model training platforms needed for next-generation AI methods, and improve infrastructure driven by next-generation AI methods.

(iii) Applications of next-generation AI methods geared towards scientific fields.

Develop new physics models (物理模型) and algorithms, build open-source scientific databases, knowledge bases, physics model libraries, and algorithm libraries, and promote exemplary applications of new AI methods in solving complex problems in scientific fields.

III. Research Directions of 2025 Funding

(i) Cultivation Projects.

Centered around core scientific problems, and guided by the overall scientific objectives, [the National Natural Science Foundation of China (NSFC)] intends to use the Cultivation Project approach to fund applicant projects (申请项目) that have a

strongly exploratory nature, a novel topic selection, and a good foundation of previous research. The research directions are as follows:

1. New architectures for neural networks and new pre-training or self-supervised learning methods.

For data such as images, videos, graphs, and flow fields, develop more efficient novel neural network architectures, and pre-training or self-supervised learning methods, and verify them on real datasets.

2. Basic theory of deep learning.

Research the function space theory and error analysis of neural networks. Research important phenomena in the nonlinear training process for neural networks, including grokking, edge of stability, loss spikes, and condensation. Analyze the effects of the loss landscape embedding structure, training hyperparameters, condensation, regularization techniques, etc., on the sample efficiency, out-of-distribution detection, generalization ability, and inference performance of architectures such as Transformer and Mamba.

3. Fundamental problems of large models.

Research the fundamental problems of multi-task, multi-data, and large models, including but not limited to the representation theory, generalization theory, and sample efficiency of large models, as well as the training stability, scaling law, emergence, and other phenomena of large models; research the curse of memory problem of large models; research large model mechanisms such as chain of thoughts, multi-step reasoning, in-context learning, and reasoning extrapolation capabilities (such as length generalization).

4. Data-centric machine learning.

Focusing on the acquisition cost and efficiency issues on the data side of large models, analyze the optimization and generalization process of data generation models, so as to accelerate generation model training and inference; focusing on the quality and quantity of training data, design diversified and high-quality data generation methods, propose efficient data selection and matching strategies, and explore ways to iteratively optimize data and models collaboratively, so as to accelerate the training of general purpose (通用) large models.

5. AI methods and theories in scientific fields.

Focusing on typical application scenarios in scientific fields, such as nuclear fusion, drug research and development, materials design, and many-body electron problems, develop methods for the integration of AI with fundamental theories and models in

physics, chemistry, biology, etc., establish the theory and methods of generalizable cross-scale AI-assisted computation, and solve typical complex forward and inverse problems.

(ii) Key Support Projects.

Centered around core scientific problems, and guided by the overall scientific objectives, [NSFC] intends to use the Key Support Project approach to fund applicant projects that have a good accumulation of previous research achievements, that can play a role in promoting the overall scientific objectives in the areas of theory and key technologies, and that have an industry-academia-research institute-user (产学研用) foundation. The research directions are as follows:

1. Inference methods that integrate logic and deep learning.

Improve the reliability of AI systems and their ability to handle complex inference problems by fusing together logic methods and deep learning methods. By constructing formal mathematical databases, optimizing machine translation technology and proof algorithms, achieve highly automated theorem proving, and establish highly rigorous inference models.

2. Geometric generation that integrates physics and AI.

Achieve a fundamental generative model for multiphysical microgeometric structures by integrating physical simulation and deep learning methods; by establishing a database of formalized and boundary representation (B-Rep)-expressed computer-aided geometric [design] (CAGD) models, achieve automatic generation of CAGD models with accurate functional descriptions and high precision, and construct high-quality geometric assets; and strengthen the robustness of tetrahedral, quadrilateral, and higher-order grids (高阶网格) generated from CAGD models to increase the stability and accuracy of physical simulations.

3. New generation brain-inspired AI.

Focusing on the energy-saving characteristics and diverse structures of biological neurons, establish a simple and efficient mapping between biological neurons and artificial neurons, so that artificial neurons have the energy optimization and dendritic nonlinear computation functions of biological neurons, and propose a unified energy optimization algorithm framework. Combining neural network connection structure with the characteristics of brain regions, design a model of an artificial neural network bound by the characteristics of biological neurons and based on spatiotemporal information embedding, so as to achieve advanced cognitive functions such as memory and decision-making. Achieve at least three kinds of mapping between biological and

artificial neurons and at least three kinds of dendritic computation functions, thereby achieving improved accuracy, performance, and explainability compared with existing mappings.

4. Humanlike cognitive learning framework.

Explore humanlike cognitive learning frameworks, improve the intelligentization level of systems through the active perception and interaction of intelligent agents (智能体), and the autonomous learning and iterative updating of the model, and solve the interactive decision-making problems of embodied intelligent agents in complex physical environments, thereby providing support for embodied AI (具身智能) tasks such as causal model construction and physical common sense generation.

5. Credible generation of multi-agent simulation scenarios driven by physical processes.

Build a scenario model driven by physical principles to achieve highly realistic dynamic simulation of the environment; integrate multimodal information to enhance the interaction between intelligent agents and the environment, and ensure that the characteristics and behaviors of simulation entities match those of the real world; and establish and improve a simulation data credibility assessment system to ensure the reliability of simulation scenarios, thereby providing a solid virtual foundation for complex systems research and decision-making.

6. Explainable AI methods and their application in complex systems of chemical reactions.

Develop explainable AI methods based on the fusion of deep learning and physics models to analyze and construct chemical reaction networks, reveal the micro-mechanisms and surface-interface activity laws of complex chemical reaction systems, and promote the practical application of explainable AI methods in important fields such as energy catalysis, synthetic chemistry, and synthetic biology.

7. AI-driven virtual cell research.

Based on multi-omics data and AI methods, develop virtual cell technology to simulate dynamic cellular processes, analyze the principles of cell interactions, and predict the complex responses of organisms to disturbances. Specifically, this includes: 1) Virtual single cells: Focusing on typical scenarios such as yeast and cell lines, construct multi-scale gene regulation and signaling pathway network models using multi-omics data to dynamically predict cell behavior, and achieve applications in synthetic biology or drug screening; and 2) Virtual organisms: Focusing on typical scenarios such as nematodes and embryos, construct models of multicellular

interactivity relationships and spatiotemporal dynamic change using imaging and omics data, so as to reveal the biological mechanisms of organisms.

8. Rare disease diagnosis and decision-making large models.

Establish a large-scale clinical and genetic information database for rare diseases, covering multidimensional information such as genes and clinical manifestations, and, combining multimodal data, construct explainable diagnostic decision-making large models for rare diseases, thereby providing a key basis for the diagnosis and treatment of rare diseases.

9. Multimodal large model-based design of biological components (生物元件) tolerant to extreme environments.

Based on extreme environment microbial data, construct a multimodal and pre-trained large model for proteins and nucleic acid sequences that integrates sequence, structure, and function; analyze the connections between important biological components and environmental adaptability, metabolic function, etc., and construct a prediction model for the adaptability of components to extreme environments; and design and develop functional components such as proteins and nucleic acids that can tolerate high temperatures, high pressures, and extreme pH, and validate them in wet labs, so as to promote their industrialized application in fields such as industry and medicine.

(iii) Integration Projects.

This year, [NSFC] intends to select research directions with significant application value and a good research basis for integrated funding, with research directions as follows:

1. General purpose large models with memory-reasoning separation and hierarchy (记忆与推理分离、分层).

Design a new architecture for a general purpose large model with memory-reasoning separation and hierarchy, construct an inference dataset (推理数据集), and research the basic theory and training methods of large models. Specifically, this includes: (1) Exploring a model architecture with memory-reasoning separation, achieving scalable, learning-enabled, highly compressible, distributed, and hierarchical memory storage, designing new training methods with high-efficiency storage and computing, and verifying the effectiveness of the new architecture and training methods by pre-training large models (no fewer than 7B parameters and 1T tokens) from scratch; 2) Achieving automatic extraction of textual inference data, and automatic construction of natural language inference data and formal mathematical theorem

databases; and 3) Clarifying the intrinsic mechanism(s) for the effectiveness of the next-token prediction training paradigm, and studying the impact of hyperparameters and model complexity on the inference ability of large models.

2. Construction methods and applications of the structure-performance relationships of structural materials.

Research knowledge codification representation methods for the composition, organizational structure, processing, etc., of structural materials, develop materials knowledge construction algorithms such as symbolic regression and deep learning, and construct mathematically expressive models or empirical models of structure-performance relationships with clear physical meanings for typical structural materials; research methods such as interpretable material feature engineering, knowledge-induced high-precision modeling, and material causal inference, uncover the intrinsic relationships in the effects of multi-component composition, complex processes, and organizational structures on material properties, and establish materials databases, knowledge bases, and technology libraries; develop a data- and knowledge-driven approach for the entire process of new structural material R&D and manufacturing, and develop 2-3 types of high-performance typical metal structural materials, verifying them through engineering trials.

3. An intelligent operating system that integrates environment, system, and model.

Focusing on the generalization and intelligentization (智能化) of computing hardware and devices in the physical world, and breaking down the boundaries between the physical environment and intelligent models, design a scheme for the coevolution of environment, system, and model. Specifically, this includes: 1) Researching a method for the joint iterative evolution of the operating system, AI model, and physical environment to assure environment-system-model coevolution, thereby improving overall performance at least 50% compared to separate evolution; 2) Researching a distributed operating system meta-architecture, geared towards diverse computing hardware and physical devices, that supports the efficient abstraction and adaptation of at least five types of hardware and devices, and improves utilization efficiency by at least 30% compared to separate abstraction; and 3) Researching a highly reliable distributed data transmission and storage platform, geared towards complex physical environments, that achieves the resilient interconnection and real-time collaboration of intelligent elements (智能元素) in physical space.

IV. Basic Principles for Project Selection

- (i) Closely centered on the core scientific problems, cutting-edge exploration with strong foundational and interdisciplinary characteristics will be encouraged; priority will be given to supporting original research.
- (ii) Priority will be given to research projects that develop new methods for next-generation AI or promote the application of new AI methods in scientific fields.
- (iii) Key support projects and integration projects should have a good research basis and accumulation of previous [research achievements], and directly contribute to and support the overall scientific objectives.

V. 2025 Funding Plan

[NSFC] plans to fund about 15 cultivation projects, with funding for direct costs of around Chinese yuan Renminbi (RMB) 400,000 per project and a funding period of 3 years. Application forms for cultivation projects should indicate “January 1, 2026 – December 31, 2028” as the research period. [NFSC] plans to fund about 6 key support projects, with funding for direct costs of around RMB 3 million per project and a funding period of 4 years. Application forms for key support projects should indicate “January 1, 2026 – December 31, 2029” as the research period. [NSFC] plans to fund about 3 integration projects, with a funding for direct costs of RMB 8-12 million per project and a funding period of 4 years. Application forms for integration projects should indicate “January 1, 2026 – December 31, 2029” as the research period.

VI. Application Requirements and Matters of Importance

- (i) Applicant eligibility.

Applicants for projects in the Major Research Program should meet the following conditions:

1. Have experience in undertaking basic research projects;
2. Hold a senior professional technical position (title).

Postdoctoral researchers, individuals currently pursuing graduate degrees, and individuals who do not have a work unit (工作单位) or whose unit is not a [NFSC-registered] host unit (依托单位) are not eligible to apply.

- (ii) Application limit provisions.

The relevant requirements of the application limit provisions in the “Application Provisions” of the “2025 National Natural Science Foundation of China Project Guide” shall be implemented.

- (iii) Matters of importance for applying.

Applicants and host units should carefully read and implement the relevant requirements of this project guide, the *National Natural Science Foundation of China Guide to Programs 2025* and the *Notice on Application and Final Findings of National Natural Science Foundation Projects in 2025*.

1. Paperless applications will be implemented for the projects of the Major Research Program. The submission period for application forms is from March 1, 2025 to 16:00 on March 20, 2025.

- (a) Applicants should fill out and submit electronic application forms and attachments online according to the Major Research Program project filing instructions and drafting outlines in the NSFC online information system.
- (b) The Major Research Program aims to strategically provide directional guidance and integration of strengths for related multidisciplinary research, closely centered on core scientific issues, forming project clusters. Applicants should independently draft project titles, scientific objectives, research content, technical routes, associated research expenses, etc. according to the core scientific questions proposed to be solved by the Major Research Program and the proposed funding research directions in the project guide.
- (c) In the application form, select “Major Research Program” as the funding category; select “Cultivation Project,” “Key Support Project,” or “Integration Project” as the subcategory; select “Explainable and Generalizable Next-Generation AI Methods” in the notes field; and select T01 as the handling code. Select no more than five application codes according to the specific research content.

Cultivation Projects and Key Support Projects may not have more than two cooperating research units (研究单位), and Integration Projects may not have more than four cooperating research units. The main participants in an Integration Project must be actual contributors to the project, and the total number of participants must not exceed 9.

- (d) In the initial part of the application, the applicant should clearly state the funding research directions in the project guide to which the application conforms, as well the contributions that would be made towards solving the core scientific problems of the Major Research Program and achieving the Major Research Program’s scientific objectives.

If the applicant has already participated in other S&T programs and projects related to the Major Research Program, they should discuss the differences and

connections between the applicant project and [their] other related projects in the “Research Basis and Work Qualifications” section of the main text of the application.

2. The host unit should complete the host unit letter of commitment, organize the application, and review the application materials, according to the requirements. [The host unit] should verify each item of its electronic application and attached materials through the [NSFC] information system before 16:00 on March 20, 2025, and submit the unit’s project application checklist online before 16:00 on March 21.

3. Other matters of importance.

(a) In order to achieve the Major Research Program’s overall scientific objectives and multidisciplinary integration, project principals (项目负责人) of funded research projects should commit to abiding by the relevant data and information management and sharing regulations. During the process of project implementation, [project principals] should pay attention to the mutually supportive relationships with other projects in the Major Research Program.

- (b) In order to strengthen projects’ academic exchange, and promote project cluster formation and interdisciplinary intersection and integration, the Major Research Program will host an annual academic exchange conference for funded projects, and will organize academic seminars for related fields from time to time. Project principals of funded projects are obligated to participate in the aforementioned academic exchange activities organized by the Expert Steering Group and Management Work Group of the Major Research Program.

- (iv) Contact information.

First Division for Interdisciplinary Science, Department of Interdisciplinary Science
Contact phone number: 010-62328382